The cost of self-sufficiency: Robust energy system planning under prolonged transmission blackouts

ONGOING WORK

ECEMP 2026 — Session on 'Energy security and geopolitical risks'

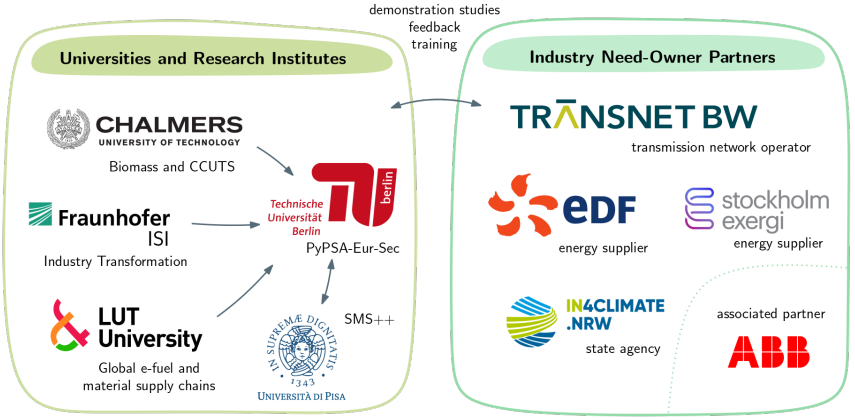
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Supported by:
Federal Ministry for Economic Affairs and Energy
on the basis of a decision by the German Bundestag

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RESILIENT project partners



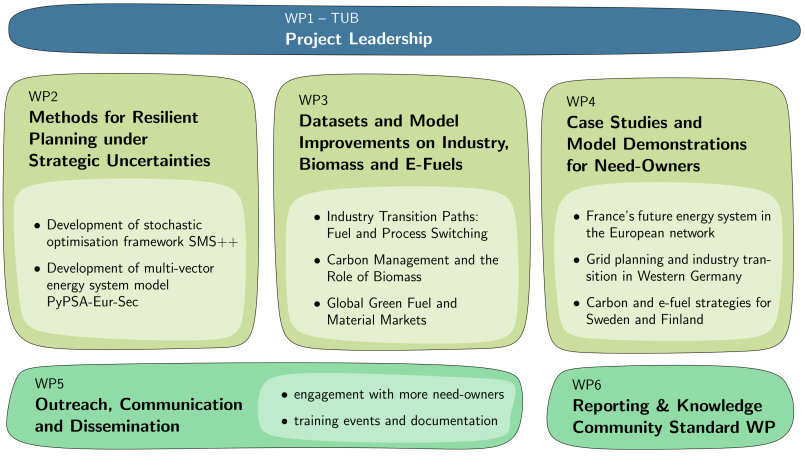
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Funded through the **CETPartnership 2022** call and by the **German Federal Ministry for Economic Affairs and Energy (BMWE)** for the German project partners.

Source: https://resilient-project.github.io/

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Working packages



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Source: https://resilient-project.github.io/

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Motivation: the grid is no longer a given

Recent outages and attacks in Europe

- **Ukraine, since 2022:** systematic strikes on critical infrastructure, incl. generation and grid
- **Baltic Sea, Oct 2023 & Dec 2024:** Balticconnector gas pipeline and Estlink 2 power cable damaged by dragged ship anchors; Estlink 2 out ∼6 months
- **Iberia, Apr 2025:** whole peninsula blacked out, ∼10 h to restore most areas
- **Berlin, Sep 2025 & Jan 2026:** arson on high-voltage cables, ∼50 k / 45 k households dark for 2.5 / 4 days

What current planning models do

- Resilience studied for **weather extremes**, Dunkelflaute events, and **fuel-import disruption**
- $N-1$ covers the instant loss of *one* element
- The grid itself is assumed **always available**

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What if every region had to ride through on its own → for hours, days, or weeks?

Source: ENTSO-E/ACER expert panel (2025); IEA (2026) *Energy System Resilience – Lessons from Ukraine*; Stromnetz Berlin (2025, 2026); Finnish NBI (2023, 2024); Fingrid (2025); Directive (EU) 2022/2557. Weather: Grochowicz et al. (2025), Bernecker et al. (2025). Imports: Riepin et al. (2022), Keutz & Kopp (2025), Göke et al. (2025).

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What does it take for every region to ride through on its own?

1. How much **more** must a net-zero German power and electrified-heat system **invest** so that each region can serve its own load when **all inter-regional transmission fails for 6 h to 2 weeks**?
2. **Which technologies** (batteries, iron-air, hydrogen, firm capacity) **and which regions** carry that resilience, and how does the mix shift as outages get longer?
3. Can a **stored renewable methanol reserve** contribute to resilience, and how much does it **impact the cost of long outages**?

Why adaptive robust optimisation?

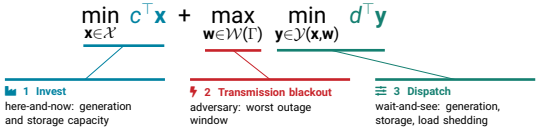
- No credible probability for *when* an attack hits → plan against the **worst start time**
- Investment fixed before; dispatch **and storage** adapt during the outage
- Worst window searched **across the whole weather year**

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Source: ARO: Bertsimas et al. (2013); Zeng & Zhao (2013); applied to energy planning in Riepin et al. (2022), Bernecker et al. (2025).

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Method: adaptive robust optimisation (ARO)



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Method: adaptive robust optimisation (ARO)

$$\min_{\mathbf{x}\in\mathcal{X}} \; c^\top \mathbf{x} \;+\; \max_{\mathbf{w}\in\mathcal{W}(\Gamma)} \; \min_{\mathbf{y}\in\mathcal{Y}(\mathbf{x},\mathbf{w})} \; d^\top \mathbf{y}$$

1 Invest
here-and-now: generation and storage capacity

2 Transmission blackout
adversary: worst outage window

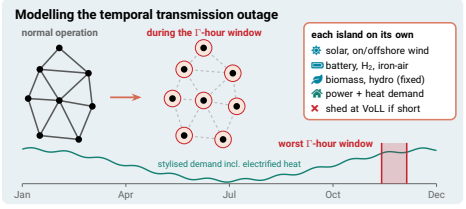
3 Dispatch
wait-and-see: generation, storage, load shedding

Uncertainty set

$\mathcal{W}(\Gamma) = \{\mathbf{w} \in \{0,1\}^T : \; w_t = 1 \text{ on one contiguous block of } \Gamma \text{ hours}\}$

$w_t = 1 \;\Rightarrow$ **all** AC lines and DC links are out: every region becomes an island and must ride through on local generation and storage.

Capacity is committed before the timing is known; dispatch adapts once it is. No probabilities needed: the worst window is found endogenously.



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Method: adaptive robust optimisation (ARO)

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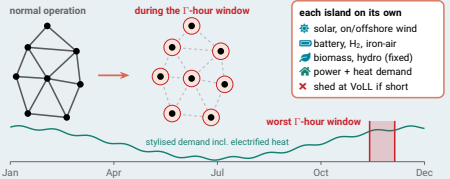
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Modelling the temporal transmission outage



Solving the ARO problem via Column-and-Constraint Generation (C&CG)

Master (LP)
$\min_{\mathbf{x}} \; c^\top \mathbf{x} + \max_{k\in\mathcal{K}} \text{opex}_k(\mathbf{x})$
one dispatch per window found so far; CVaR with ω=1 makes the max exact/robust
$\Rightarrow$ **lower bound**

design $\mathbf{x}^*$

worst window $\mathbf{w}^*$

Subproblem (worst window)
$\max_{\mathbf{w}\in\mathcal{W}(\Gamma)} \min_{\mathbf{y}} \; d^\top \mathbf{y}$ given $\mathbf{x}^*$
exact enumeration: one LP per start t_0, in parallel (alternatively, a dualised single MILP)
$\Rightarrow$ **upper bound**

repeat until $(\text{UB} - \text{LB})/\text{UB} < \epsilon = 10^{-4}$

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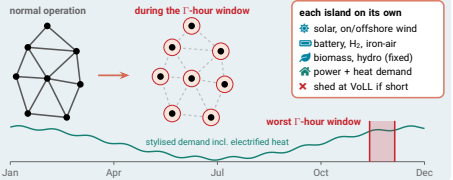
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Case study (Germany)

Network
PyPSA-Eur Germany, 8 (arbitrary) nodes, 2050

Time
full year, 3-hourly, weather year 2013

Heat
electrified: heat pumps + resistive, exogenous load

Invest
solar, wind, battery, H_2, iron-air; + methanol reserve (scenario)

Policy
zero CO_2: no fossil backup, no imports from neighbours

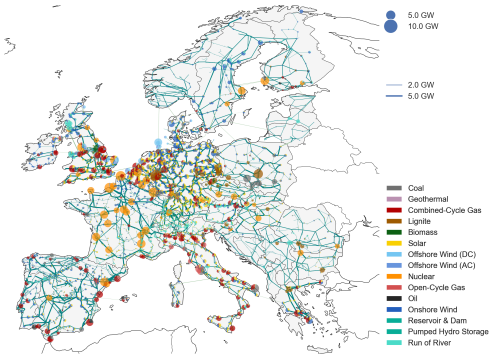
Sweep
$\Gamma \in \{6, 24, 72, 168, 336\}$ h
VoLL 10 000 €/MWh

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PyPSA-Eur: A sector-coupled energy system model for Europe

- Spatially and temporally highly resolved linear optimisation model covering the **European continent**,
- Based on the open-source framework **PyPSA**,
- Includes **existing power plant fleet**, **renewable potentials**, and **availability time series**,
- Includes the **electrical transmission grid** from 220 kV to 750 kV (AC, including Ukraine) and from 150 kV (DC), with the option to enable planned grid expansion projects (TYNDP and German NEP), as well as the gas network,
- Maintained by the Department of Digital Transformation in Energy Systems at **TU Berlin**.



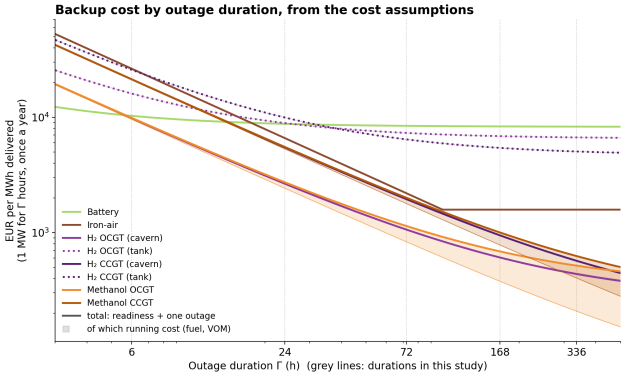
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Source: https://pypsa-eur.readthedocs.io

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Two scenarios: storage only, or storage plus a methanol reserve

- **No methanol:** battery, iron-air, H_2 (caverns where geology allows + local tanks)
- **Methanol:** additionally buy renewable methanol at 130 €/MWh, store it in local tanks, burn it in turbines *only during an outage*
- Holding fuel in tanks is cheap; turbine capacity is not. Long outages favour **cheap energy, expensive power**



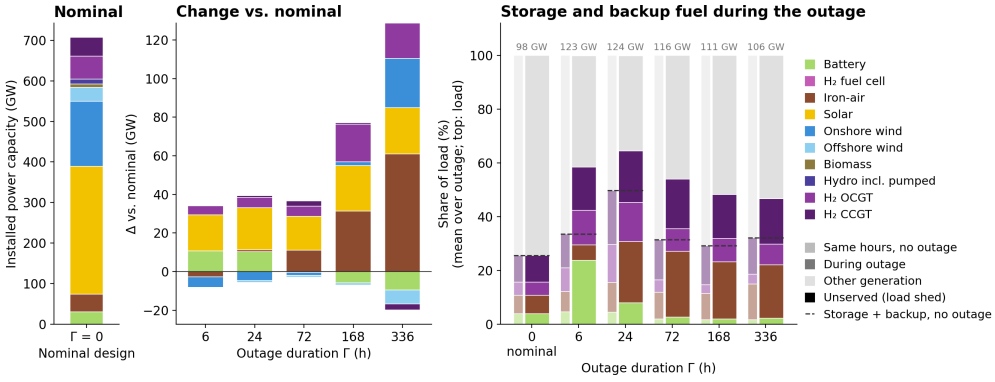
Standalone sizing for 1 MW over Γ hours, outage once a year. Running cost = purchased methanol + VOM. **Not included:** electricity to recharge storage or make H_2 (round-trip losses, electrolyser), shared with normal operation and paid in the optimisation.

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Source: Costs from the model's cost assumptions. Methanol price: Wenzel et al. (2026); minimal methanol economy: Glaum et al. (2025).

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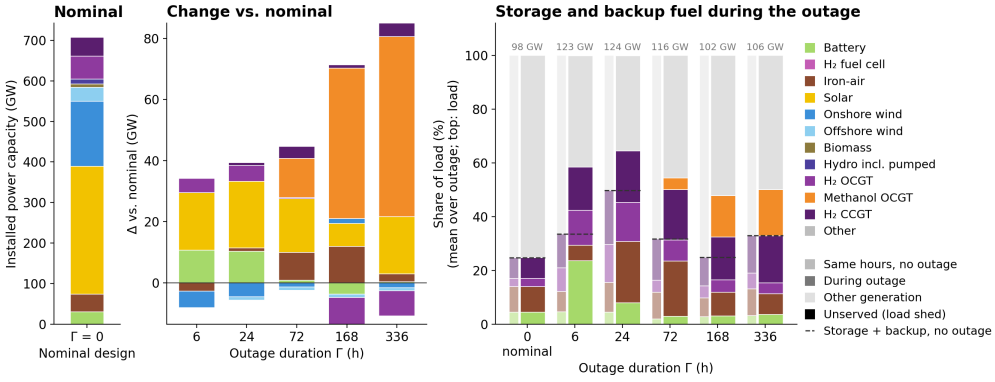
Storage can secure multi-day autonomy for +0.3–12.4 %



Batteries cover hours, **iron-air** takes over beyond 3 days (+61 GW at 336 h); from there, each extra day of autonomy adds a steady ∼0.7 bn€/a.

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Methanol-based backup makes long outages cheaper to survive



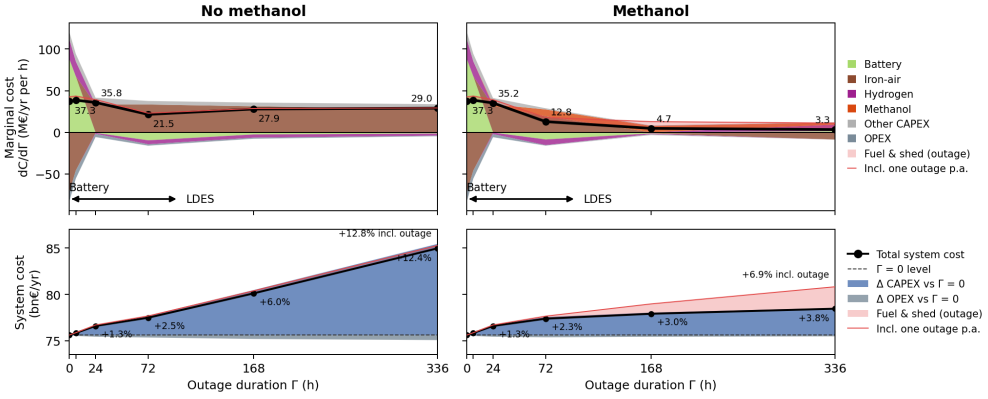
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Two weeks of autonomy for **+3.8 %** instead of +12.4 % with storage alone. Identical up to 24 h; from 72 h on, methanol turbines replace iron-air and each extra hour only adds fuel to the tank (19 TWh at 336 h).

Source: Costs: all installed capex + no-outage opex; fuel burned during the event excluded. 336 h is the best design after 9 C&CG iterations (gap 0.08 % of total cost); all other points converged.

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What one more hour of autonomy costs

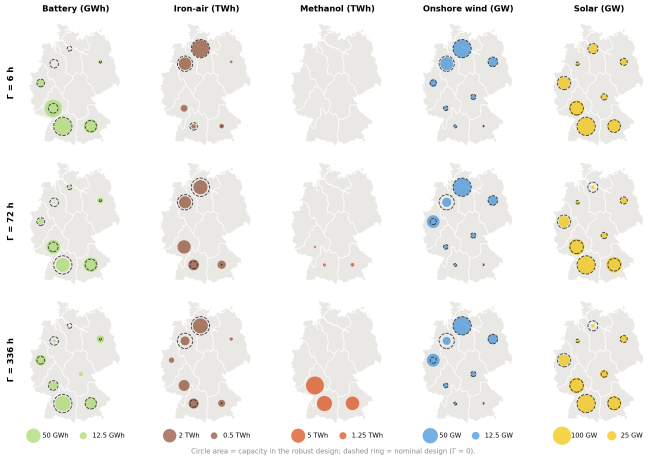


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First day: ∼37 M€/a per outage hour (batteries). Beyond 3 days: ∼28 with storage alone (iron-air) vs. 3–5 with methanol (∼11 incl. one outage's fuel a year), i.e. each extra day costs ∼**0.7** vs. **0.1–0.3 bn€/a**.

Source: Top: $dC/d\Gamma$ at each modelled Γ (central differences). Bottom: total system cost (all installed capex + no-outage opex). Red: fuel & shedding if the outage happens once a year.

Islands need local supply, resilience moves south



German case study:

Onshore wind leaves the Northwest; solar shifts from the north to Hesse and Bavaria.

Batteries move south for short outages only.

The methanol reserve sits only in the south, where there are no salt caverns for H_2.

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Limitation: the hedge pays off if storage is held in reserve

	capex premium	payback = avoided damage / premium	
Γ	bn €/a	with foresight	**no foresight**
6 h	0.29	8×	**1.1**×
24 h	1.14	10×	**1.6**×
no methanol / methanol			
72 h	2.1 / 2.0	15 / 16×	**5.5 / 8.1**×
168 h	4.9 / 2.5	15 / 29×	**6.4 / 22**×
336 h	9.9 / 3.0	15 / 48×	**8.0 / 36**×

Robust design vs. the non-robust design, re-evaluated over all outage windows with storage levels fixed at the outage start.

- The model sizes and **pre-charges storage** knowing when the outage starts
- Short Γ: without foresight the premium barely pays back (1.1–1.6×), but it is also small (0.3–1.1 bn€/a), so little is at stake
- Long Γ: robustness **survives**, still 5–36× payback
- A held **fuel reserve is more resilient**, e.g. the methanol design is the most foresight-proof

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Main findings

1. **Up to 3 days, resilience is cheap:** +0.3 % (6 h), +1.3 % (24 h), +2.3–2.5 % (72 h) of system cost — 0.2 to 1.9 bn€/a.
2. **Beyond 3 days, the hedge decides the bill:** two weeks cost **+12.4 %** (9.4 bn€/a) with storage alone, **+3.8 %** (2.9 bn€/a) with a methanol reserve — 5.2 bn€/a if the fuel of one outage per year is counted.
3. **The cheapest hedge changes with duration:** batteries for hours, iron-air for days, a fuel stock for weeks. The stock scales with turbine capacity $\times$ Γ (19 TWh at 2 weeks), so each extra hour only adds fuel.
4. **Resilience is local:** islands need supply where the load is, in the case of Germany, onshore wind leaves the Northwest, solar and the methanol reserve go south.
5. **Storage levels at the outage start matter:** with sufficient reserves held, robust designs pay back many times over for outages beyond 3 days, most of all with the methanol reserve.

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Source: Preliminary: German 8-region case, one weather year; costs exclude fuel burned during the outage.

What this means for European (German) energy policy

1. **Set an explicit resilience target:** name the outage duration to plan for. The cost curve prices each extra day.
2. **Regulate reserves, do not leave them (fully) to the market:** resilient capacity only pays off if it is held in reserve, i.e. a storage obligation, like the EU's 90-day oil stocks.
3. **Recognise strategic fuel reserves,** e.g. methanol: the cheapest hedge beyond ∼3 days where there are no H_2 caverns
4. **Plan resilience regionally:** firm capacity near load, alternatives where there are no H_2 caverns.
5. **Stress-test grid loss** in adequacy assessments (ERAA, TYNDP), next to weather extremes.

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Source: Preliminary: German 8-region case, one weather year. Oil stocks: Council Directive 2009/119/EC. Risk assessments of critical entities: Directive (EU) 2022/2557.

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Outlook

- **Scale to Europe:** 156 NUTS1/2 regions; each country plans against its own worst case, border flows from a European pre-run
- **Several weather years**, VoLL sensitivity, finer regions
- Cost sensitivities on methanol
- (**Future work outside this scope**) **Non-anticipative recourse:** stays two-stage, because only the outage *onset* is uncertain

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How much resilience does Europe need, and how much does it cost?

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Thank you.

The cost of self-sufficiency
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Backup: caveats

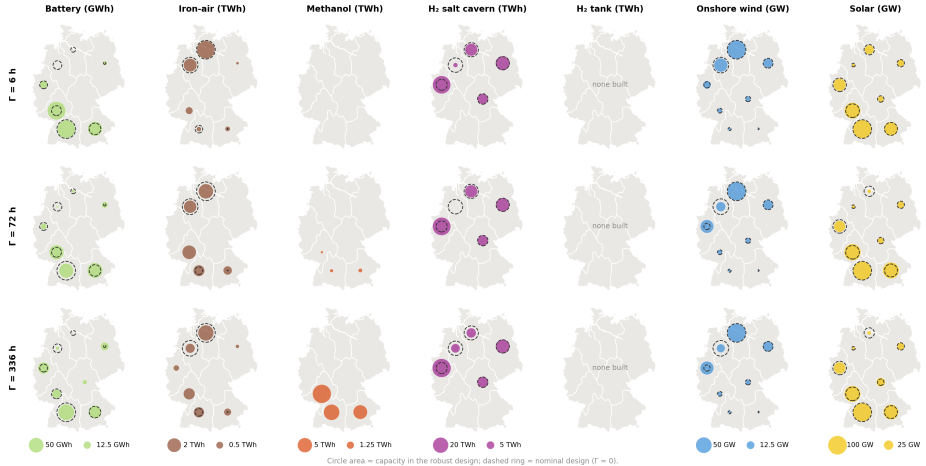
- **Controlled islanding**, not black start: regions stay energised; restart after a collapse is out of scope and strictly worse
- 8 regions: islands far larger than reality $\Rightarrow$ premium likely a **lower bound**
- One weather year (2013), 3-hourly; no imports from neighbours
- VoLL 10 000 €/MWh; heat as an inflexible load
- Reported costs exclude the outage's own running cost. Counting the fuel burned in one outage per year shrinks methanol's advantage at 2 weeks from 3.3$\times$ to $\sim$1.9$\times$ (premium 5.2 vs 9.7 bn€/a; red on the marginal-cost slide)

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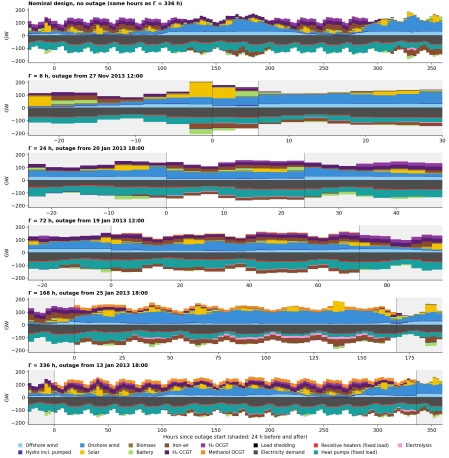
Backup: regional capacity incl. H_2 storage



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Backup: dispatch in the worst window (methanol)



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Backup: references

- Bernecker et al. (2025). Adaptive robust optimization for European electricity system planning considering regional Dunkelflaute events. SSRN 5361178.
- Bertsimas et al. (2013). Adaptive robust optimization for the security constrained unit commitment problem. *IEEE Trans. Power Syst.* 28(1), 52–63.
- Council Directive 2009/119/EC on maintaining minimum stocks of crude oil and/or petroleum products.
- Directive (EU) 2022/2557 on the resilience of critical entities.
- Expert Panel on the Grid Incident on 28 April 2025, ACER, ENTSO-E (2025). Grid incident in Spain and Portugal on 28 April 2025 – Factual report.
- Fingrid (2025). EstLink 2 electricity transmission link returns to commercial use, press release of 18 Jun 2025.
- Finnish National Bureau of Investigation (2023, 2024). Balticconnector and Estlink 2 investigations, press statements of Oct–Nov 2023 and 26–30 Dec 2024.
- Glaum, Neumann, Millinger, Brown (2025). The minimal methanol economy as a gap-filler for high electrification scenarios. arXiv:2505.09277.
- Göke, Wohland, Moret, Bardow (2025). The liquid buffer: multi-year storage for defossilization and energy security under climate uncertainty. arXiv:2511.13513.
- Grochowicz, Bloomfield, Victoria (2025). Preparing for the worst: long-term and short-term weather extremes in resource adequacy assessment. arXiv:2508.05163.
- International Energy Agency (2026). Energy system resilience – Lessons learned from Ukraine.
- Keutz, Kopp (2025). The cost of security: analyzing strategies to hedge hydrogen import disruption under stochastic representation of weather. EWI.
- Riepin, Schmidt, Baringo, Müsgens (2022). Adaptive robust optimization for European strategic gas infrastructure planning. *Appl. Energy* 324, 119686.
- Stromnetz Berlin (2025, 2026). Press releases on the Johannisthal (9–11 Sep 2025) and Lichterfelde (3–7 Jan 2026) outages.
- Wenzel et al. (2026). Direct air capture enables sustainable methanol production in water-scarce regions. *Nature Communications* 17, 8495.
- Zeng, Zhao (2013). Solving two-stage robust optimization problems using a column-and-constraint generation method. *Oper. Res. Lett.* 41(5), 457–461.

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